Sharper One-Sample Matroid Prophet Inequalities

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Written proof with same-agent verification; independent mathematical review and formalization pending.

Conditional activation safety, fixed activation scores, and joint mask membership raise the guarantee above 2^{-31}. The paper states the construction and source interfaces, proves the shared lemmas, and derives the selected coefficient below.

1 Current theorem and dependencies

For every known finite labeled matroid M on n elements, there is a measurable, distribution-independent online rule using one independent sample per label and $40n + 26$ initial independent fair bits. All sample/value coordinates are mutually independent and nonnegative, with matching laws at each label, and the seed is independent. Assume only that $\mathbb{E}[\mathrm{OPT}_M(V)] < \infty$. For every measurable arrival permutation depending on all samples, all values, and the entire seed, the accepted set is independent after every prefix and

$$\mathbb{E}[\text{reward}] \geq C_\star \mathbb{E}[\mathrm{OPT}_M(V)], \qquad C_\star = \frac{8003458533570087953}{168074759461507046572032000000} > \frac{51}{50}\,2^{-31}.$$

The exact certificate is approximately **1.14973 times** the preceding `(7/4)*2^-32` certificate (**14.97% higher**). This compares guaranteed fractions, not measured performance.

We retain the density paths, generators, hybrid safety comparison, marked-pivot machinery, rank accounting, and lossless simulation from OpenAI's pinned manuscript. Their interfaces are specified below. Computation and storage may be exponential, with exact observations; no polynomial-time claim is made.

2 The fixed-vector rule and its statistics

Fix a nonnegative hidden vector w before any seed bits. The rule reveals and sacrifices a seed-chosen mask and must handle every permutation; its target is $\mathbb{E}_R \min_\pi w(A(w, R, \pi))$. Write $\mathrm{r}(A \mid P) = \mathrm{r}(A \cup P) - \mathrm{r}(P)$ and $F_\varnothing = \mathrm{cl}(\varnothing)$. Fix a label order for every tie. Higher rounded weight means higher priority. Put $W = \mathrm{OPT}_M(u)$ and let I^* be priority greedy on the full rounded vector.

Let $\kappa \geq 4$, $0 < h, \rho, t < 1$, integer $L \geq 1$, and $\theta > 1$ be fixed parameters supplied by the selected result. The independent masks H, D, C, T have rates h, ρ, t, t respectively. The main branch has probability $1 - t$; parity is fair. Choose a uniform shared phase $J \in \{0, \ldots, L-1\}$, set $B = \theta^L$, and define

$$u_J(x) = \theta^J B^{\lfloor \log_B(x/\theta^J) \rfloor} \quad (x > 0), \qquad u_J(0) = 0.$$

All random choices are made initially. Each result supplies exact dyadic mask probabilities and a finite fair-bit implementation. The score variables in the counting proof below are analysis devices, not algorithmic inputs.

The maximum branch reveals a separate fair mask and accepts the first unmasked positive nonloop with priority above every revealed positive nonloop. An empty threshold is below every positive nonloop. If m_* is the largest rounded nonloop weight, expected worst-order reward is at least $m_*/4$: the largest-priority nonloop is unmasked and the second is masked with probability $1/4$. With only one positive nonloop success has probability $1/2$; with none the bound is zero.

In the main branch reveal and sacrifice $H \cup D \cup C$. Define

$$U = \{e \notin H : u_e > 0,\ e \notin \mathrm{cl}(H_{>e})\}, \qquad Y = I^* \setminus H.$$

The candidate test uses only revealed H values and the tested value. For a fixed phase divide weights by θ^J , so true groups U_i have weight B^i . List those meeting D in increasing weight order as $G_1, \dots, G_m$; put $D_j = D \cap G_j$, and similarly C_j, T_j . The full groups are analysis objects; their revealed D_j, C_j parts suffice for preparation. If there are no listed groups, reject all.

For a flat P , let $Q_j(P)$ be the largest flat $Z \supseteq P$ maximizing $|D_j \cap Z| - \kappa\, \mathrm{r}(Z)$. Take closure first for nonflat arguments. Set $a_j = -3j$ and enable $Q_{j,k} = Q_j$ for $k \geq a_j$, with the identity earlier. Starting from $F_0(k) = F_\varnothing$ for $k < 0$ and E otherwise, define

$$X_j(k) = Q_{j,k}(F_{j-1}(k)), \qquad F_j(k) = \mathrm{cl}(X_j(k) \cup (C_j \cap X_j(k+1))).$$

Compute paths in increasing j on the finite relevant auxiliary timeline, and put $\widehat{F} = F_m$. For a candidate nonloop in listed group j , write $b_j(e) = \min\{k : e \in X_j(k)\}$. It is eligible precisely when it is unsacrificed, T -positive, has birth $b \geq a_j + 2$ of the chosen parity, and lies outside $\widehat{F}(b-2)$. At endpoint b maintain a set A_b ; accept an eligible e exactly when $\mathrm{r}(\{e\} \mid \widehat{F}(b-2) \cup A_b) = 1$.

The source density and path lemmas give

$$F_{j-1}(k) \subseteq X_j(k) \subseteq F_j(k) \subseteq X_j(k+1), \qquad |D_j \cap (\mathrm{cl}(Q_j(P) \cup Z) \setminus Q_j(P))| \leq \kappa\, \mathrm{r}(Z \mid Q_j(P)).$$

Nonloops are born at a_j or in $[a_j + 2, 0]$, and the entire nominal path uses a generator set of size at most $|D_j|/\kappa$. The nested layer bases give prefix independence. Finite rank operations, fixed tie rules, and countably many rounding levels give measurability. These structural facts are inherited from the cited source.

For valid selected-parity endpoints $a_j + 2 \leq b \leq 0$, define

$$\begin{aligned}
K_{j,b} &= \bigcup_{\ell<j} (T_\ell \cap (X_\ell(b) \setminus X_\ell(b-1))),\\
P_{j,b} &= G_j \cap (X_j(b) \setminus X_j(b-1)), \quad O = D \cup C, \quad S_{j,b} = P_{j,b} \setminus O,\\
z_j &= \sum_b \mathrm{r}(S_{j,b} \mid X_j(b-2) \cup K_{j,b}),\\
\lambda_j &= \sum_b \mathrm{r}(T_j \cap S_{j,b} \mid \widehat{F}(b-2) \cup K_{j,b}).
\end{aligned}$$

The safe count σ_j counts $d \in D_j \cap Y$ with such a birth and $d \notin Q_j(F_{j-1}(b-2) \cup K_{j,b})$. For each nominal summand choose a fixed-order basis extension Z_b , so $\sum_b |Z_b| = z_j$. The residual is

$$\mathcal{R}_j = \bigcup_b (P_{j,b} \setminus \mathrm{cl}(X_j(b-2) \cup K_{j,b} \cup Z_b)) \subseteq O.$$

Extend σ_i, z_i, λ_i by zero on unlisted true groups. The source density argument gives $\sigma_j \le \kappa z_j + |\mathcal{R}_j|$. Its order-uniform lemma bounds the number of accepts of weight at least that of G_j below by λ_j , for every seed and order. One later-path generator budget over the whole timeline is

$$c_j = |C_j| + \sum_{\ell > j} (|C_\ell| + |D_\ell|/\kappa).$$

Conditional on H, D, C , parity, and test bits outside the focal group, independent test thinning gives $\mathbb{E}[\lambda_j \mid \mathcal{F}_{-j}] \ge t(z_j - c_j)$. These are the source interfaces used below.

3 A quadratic transaction inequality

Quadratic transaction lemma. Let $\mathcal{A}$ be an increasing event on finitely many independent bits, and let a predictable query procedure, reading each bit at most once, determine whether $\mathcal{A}$ occurs. Write $p = \mathbb{P}(\mathcal{A}^c)$. At the completed transcript let q be the failure probability with only unqueried bits random and every queried bit reset to zero. Then

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] \ge p(1-p).$$

Proof. At an intermediate transcript, let W be its remaining coordinates and I its queried positives. Let m be the actual conditional failure probability, and q the failure probability with only W random and all other bits zero. Predictable product exposure identifies m with the probability when I is forced positive and W is random. Increasingness gives $0 \le m \le q \le 1$. Initially $m = q = p$.

For a predictably chosen next coordinate, the depleted probability q' is determined before the answer and satisfies $q' \ge q$. The actual conditional failure probability is a martingale: $\mathbb{E}[m' \mid \text{transcript}] = m$. Therefore the potential $\Phi(q, m) = q^2 - 2qm$ satisfies

$$\mathbb{E}[\Phi(q', m') \mid \text{transcript}] - \Phi(q, m) = (q' - q)(q' + q - 2m) \ge 0.$$

At the end $m = \mathbf{1}_{\mathcal{A}^c}$, and failure forces $q = 1$. The initial potential is $-p^2$; its final expectation is $\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] - p$. Sum the displayed query identity over the finite number of queries, padding stopped paths by zeros, to obtain the result. The proof includes $p = 0, 1$ and transactions needing no queries.

4 The reverse exit hazard

Fix the source's filter and density masks, a listed-group prefix, and a nonloop d still alive before a reverse step. Its current states are S , and its transition is $\mathcal{G}_k(c, S)$. The remaining guard coordinates have their product law. Let $\mathcal{A}$ be the increasing event that d survives, and let p be the conditional exit probability.

For completeness, for $k < 0$ the source reverse map on a listed-group prefix is defined recursively by

$$A_0 = F_\varnothing, \qquad N_j = Q_{j,k}(A_{j-1}), \qquad A_j = \mathrm{cl}(N_j \cup \{e \in G_j \cap S_j : c_e = 1\}),$$

with output $(N_j)_j$. Starting from $X_j(0) = E$, reverse steps compute $X(k) = \mathcal{G}_k(C, X(k+1))$. For actual later states, $N_j \subseteq S_j$ for every fresh bit assignment. After computing N_j , query only $G_j \cap (S_j \setminus N_j)$; omitted additions already lie in N_j . The new remaining domain is $\bigcup_j (G_j \cap N_j)$.

The source queries exactly the departing coordinates in predictable batches. Their answers determine all output states: unqueried bits are inside flats already spanned. Thus the transcript decides $\mathcal{A}$, on both survival and failure, so the quadratic transaction lemma applies.

Let S' be the actual output and V' the remaining coordinates. For each assignment c supported on V' , monotonicity in time and state gives

$$\mathcal{G}_{k-1}(c, S') \subseteq \mathcal{G}_k(c, S).$$

The right side has all just-queried coordinates zero, exactly the depleted experiment defining q . Conditional on the full post-step transcript, the bits in V' remain independent. On survival the next hazard r therefore satisfies $r \geq q$. The states are frozen in this comparison; earlier states are not recomputed after resetting bits. No order is conditioned on.

The next alive squared hazard consequently obeys

$$\mathbb{E}[\mathbf{1}_{\text{survive}} r^2] - p^2 \geq p - 2p^2.$$

Telescope this potential over the deterministic finite reverse horizon. Every nonloop exits, and if P is its hazard at actual exit, then

$$-p_{\text{start}}^2 \geq \sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i] - 2 \sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i^2] = 1 - 2\mathbb{E}P.$$

Hence

$$\boxed{\mathbb{E}P \geq (1 + p_{\text{start}}^2)/2 \geq 1/2.}$$

In particular, the source's total squared-exit mass is at least $1/2$. Its hybrid comparison still applies because guard and test rates agree. The same baseline-birth loss and fair parity give, after averaging a density mask of arbitrary rate ρ ,

$$\mathbb{E}[\sigma_i \mid H] \geq \gamma |Y_i| - \frac{\rho n_i}{2\kappa}, \qquad \gamma = \rho/4.$$

This averages before conditioning on listing, extending the statistics by zero on unlisted groups.

The bound $1/2$ is sharp for the abstract hypotheses. Take N Bernoulli-t bits and repeatedly test the OR of the remaining suffix, querying until the first positive or until that suffix is exhausted. The next hazard equals the depleted current failure probability. With $a = 1 - t$, classification by the last positive (or none) gives

$$\mathbb{E}P = a^{2N} + t \sum_{j=0}^{N-1} a^{2j} = a^{2N} + \frac{1 - a^{2N}}{2 - t}.$$

For $t = 1/m$, $N = m^2$, this tends to $1/2$. This is an abstract limitation, not a hard matroid instance or a bound on the best prophet algorithm.

5 Joint residual encoding

Fix H , parity, and all mask bits outside a true group U of size $n \geq \kappa$. Do not condition on its focal bits or listing. Its incoming path $F(k)$, competition sets K_b , index, and activation a are fixed. Write $S_d(N) = \sum_{i=0}^{d} \binom{N}{i}$. Suppose a path has density generators J and rank witnesses Z , of sizes j, z . A generator has insertion time $\tau(g) \in [a, 0]$, and

$$X(k) = \mathrm{cl}(F(k) \cup \{g \in J : \tau(g) \le k\}).$$

The source gives two marked streams. First, list $F(k)$ and then generators inserted at k , marking the block one exactly when k is a valid endpoint of the chosen parity. Second, start with the fixed $F(b_0)$ marked zero, where $b_0 \in \{a-2, a-1\}$ has that parity. At each subsequent parity endpoint b , insert K_b, Z_b with mark zero, then $F(b)$ and generators with $b-2 < \tau(g) \le b$ with mark one. End at $b_1 \in \{0,1\}$ of that parity; set $K_b = Z_b = \varnothing$ at invalid endpoints.

The second stream has intermediate span $\mathrm{cl}(X(b-2) \cup K_b \cup Z_b)$ and endpoint span $X(b)$, since $K_b \subseteq F(b)$ and $Z_b \subseteq X(b)$. The first test selects valid births. For a label passing it, the second pivot lies in the same layer and has mark one exactly on the residual. Thus the intersection of the two mark-one sets is the residual.

Joint pivot lemma. For fixed J, Z , the number of residuals obtained from these linked streams is at most $3^j 4^{2j+z} S_{j+z}(2n)$.

Proof. The source marked-pivot argument first bounds retained greedy bases of occurrences by 4^s when s specified movable occurrences are inserted into an old sequence spanning the matroid. To recall why, old-only greedy gives an ordered basis. Only at most s of its positions can be omitted after insertions: their conditional-rank drops for the full movable set sum to at most s . There are at most 2^s choices of omitted old occurrences and 2^s of retained movable occurrences. Apply this to the two streams with $s = j$ and $s = j + z$, giving 4^{2j+z} pairs of retained bases. Fix a pair. Each nonloop has a unique minimal support in each basis, by matroid submodularity, and its pivot is the last support member.

We represent both streams by the same $j + z$ variables. Label the entire ground set $0, \ldots, N-1$, put $m = N + 1$, $T = 8m$, and $u_k = (k - a + 2)T$. Give old $F(k)$ occurrences in the first stream absolute scores 2^{u_k+4m+e} , with sign equal to their fixed marks. In the second stream, the initial $F(b_0)$ has negative scores $-2^{u_{b_0}+4m+e}$; K_b has scores -2^{u_b+e} ; and $F(b)$ has scores 2^{u_b+4m+e} . These are fixed before the focal masks.

Give each generator a variable $x_g = 2^{u_{\tau(g)}+6m+g}$. Its first score is $\eta_g x_g$, positive exactly at a valid endpoint. Its second score is $d_g x_g$, where $d_g = 1$ for the chosen parity and $d_g = 2^T$ otherwise. Multiplication by 2^T moves an insertion from $b-1$ into the endpoint b block. Give each witness $e \in Z_b$ a variable $y_e = 2^{u_b+2m+e}$, used only as the score $-y_e$ in the second stream. There are only three types for a generator:

$$(\eta_g, d_g) \in \{(1,1), (-1,1), (-1,2^T)\}.$$

The middle type includes early insertions of the chosen parity that are not valid births. Fix all types, at a cost of at most 3^j choices. Order every block by label. Second-stream generators can be reordered within their endpoint block because all have mark one; this preserves pivot marks and endpoint spans, and all retained bases are already counted.

In each stream the absolute scores are distinct powers of two in occurrence order. Every score dominates the sum of its predecessors. The sign of the sum over a support therefore equals its last member's mark. For fixed bases and types these are $2n$ affine functions of the same $j + z$ variables. Every realized mark pair is a strict sign pattern. The hyperplane-region bound gives at most $S_{j+z}(2n)$ patterns, proving the claim. Ground-set size and timeline length affect coefficients, not this dimension bound.

6 From a rank estimate to the final guarantee

Earlier versions use $a_0 = 0$ and $\bar{\gamma} = \rho/4$. The current refinement keeps a fixed additive error and pays for it with the maximum branch.

The selected result must supply constants $K > 0$, $\epsilon, a_0 \geq 0$, and $0 < \bar{\gamma} \leq \rho/4$ such that, for each true group of size $n_i \geq \kappa$,

$$\mathbb{E}[z_i \mid H] \geq \frac{\bar{\gamma}|Y_i| - \epsilon n_i - a_0}{K}.$$

The source's later-path cost, including unlisted groups by domination, obeys

$$\mathbb{E}[c_i \mid H] \leq t n_i + (t + \rho/\kappa) \sum_{j>i} n_j.$$

Independent focal test thinning therefore gives

$$\mathbb{E}[\lambda_i \mid H] \geq \frac{t}{K} \left[\bar{\gamma}|Y_i| - (\epsilon + Kt) n_i - (Kt + K\rho/\kappa) \sum_{j>i} n_j - a_0 \right].$$

Both changed cost factors are essential when $K \neq \kappa$. For general filter rate h , the source filter argument gives

$$\mathbb{E}_H u(Y) = (1-h)W, \qquad \mathbb{E}_H u(U) = \frac{1-h}{h} \mathbb{E}_H u(\text{greedy}(H)) \leq \frac{1-h}{h} W.$$

To verify the identity, condition on memberships preceding each label in rounded priority. Its span test is fixed. The candidate and sample-greedy probabilities are respectively $1-h$ and h times this test; sum their weighted contributions. The comparison set Y remains independent and contained in U .

Set $\beta = (B-1)/B$ and

$$L_0 = \epsilon + Kt + \frac{Kt + K\rho/\kappa}{B-1}, \qquad s = (1-h)(\bar{\gamma} - L_0/h).$$

For every fixed seed and arrival permutation, the source order-uniform rank bound implies that the number accepted at weight at least B^i is at least λ_i . An accepted weight B^ℓ contributes at most $\sum_{i \leq \ell} B^i = B^\ell/\beta$ to this cumulative sum. Consequently

$$\min_{\pi} u(A_{\text{main}}(R, \pi)) \geq \beta \sum_{i : n_i \geq \kappa} B^i \lambda_i.$$

The weighted higher-group cost is at most $u(U)/(B-1)$. Discarding groups smaller than κ loses comparison mass at most $\kappa m_*/\beta$, where m_* is the largest rounded nonloop weight. The geometric sum of the weights of analyzed groups is at most m_*/β, so their additive errors contribute at most $t a_0 m_*/K$. Applying the filter identities yields

$$\mathbb{E}_R \min_{\pi} \text{reward}_{\text{main}} \geq \frac{\beta t s}{K} W - \frac{t(\bar{\gamma}\kappa + a_0)}{K} m_*.$$

If $(1-t)t(\bar{\gamma}\kappa + a_0)/K \leq t/4$, the maximum branch cancels the negative term. This condition is checked by the selected result's exact certificate. For positive x , let $k = \lfloor \log_\theta x \rfloor$. Across the L phases its rounded values are θ^{k-r} , once each for $r = 0, \ldots, L-1$. Since $\theta^k \geq x/\theta$, the geometric sum gives

$$\mathbb{E}_J u_J(x) \geq \frac{\beta}{L(\theta-1)} x.$$

Apply this to one fixed true optimal independent set to obtain the same factor for the rounded optimum. Thus, provided $s > 0$, the coefficient is

$$C = \frac{(1-t)\beta^2 ts}{L(\theta-1)K}.$$

The minimum over permutations remains inside the seed expectation at every stage, including phase and branch. The source's glued-vector simulation reveals and sacrifices the mask fixed by the seed, using the sample there and the online value elsewhere. Conditional on the entire seed this glued vector has the original product value law, so it is independent of the seed. Ignored samples, sacrificed values, and unused coins may still inform the adversary's chosen permutation; the pointwise minimum covers them all.

7 Resources and proof dependencies

The rule uses a known finite labeled matroid, independent nonnegative values and matching independent samples, and exact observations. Only the expected offline optimum needs to be finite. Rank/closure tables and flat enumeration may be exponential; preprocessing and storage are part of the construction. There are finitely many occupied groups and a finite relevant auxiliary timeline. Observation precision and integer level indices are unbounded in the original measurable model. For rational inputs, exact power comparisons have costs depending on bit lengths and exponents. No polynomial complexity or practical performance claim is made.

The source density, generator, path, feasibility, hybrid safety, thinning, order-uniform rank, and simulation lemmas remain explicit dependencies. The quadratic and joint pivot proofs above are this project's written refinements. Exact certificates prove their scalar assertions, and finite checks test their specified cases. Independent mathematical review and formalization of these refinements remain pending.

8 Activation generators and joint mask membership

The shared framework supplies the rule, quadratic exit bound, linked streams, and final payoff calculation. We retain the preceding reconstruction and projection lemmas, recalled here in full, then improve three linked estimates: safety conditional on activation membership, the fixed scores of activation generators, and the joint mask-membership probability. The comparison optimum and new classifications are analysis objects; the online rule uses neither.

8.1 Retained insertions determine the old basis

Reconstruction lemma. Fix an ordered list of old occurrences in a finite matroid and interleave specified movable occurrences arbitrarily. If the greedy scan retains movable occurrence set T, its retained old occurrences are exactly the greedy scan of the old list after contracting the labels of T. Thus s specified movable occurrences admit at most 2^s retained bases of occurrences.

Proof. Let A be the retained old occurrences preceding an old occurrence e. If the interleaved scan rejects e, its label is spanned by A and the movable occurrences already retained, a subset of T. It is therefore spanned by $A \cup T$. If the scan retains e, the final retained set is independent and

contains $A \cup T \cup \{e\}$, so e increases rank over $A \cup T$. Induction over the old list proves the contraction-scan identity. Occurrences are distinguished even when labels repeat; retained occurrences cannot repeat a label or be loops. The argument includes $T = \varnothing$. For fixed T the whole retained basis is fixed, and there are at most 2^s choices of T.

In particular, the source's general marked-pivot bound improves from $4^s S_s(m)$ to $2^s S_s(m)$ with its affine-sign proof unchanged. For the actual two streams, forced retention gives a stronger bound still.

Choose the density generators by the source's canonical forward extension rule. At time k, the first-stream prefix before the new generators spans $\mathrm{cl}(F(k) \cup X(k-1))$. Each new generator increases rank over this flat and earlier generators in its batch. Every generator is therefore retained in that stream, so its retained basis has one possibility for fixed generator set J.

Every witness in Z_b is retained in the second stream: its prefix spans $X(b-2) \cup K_b$, and Z_b is an independent extension over that base. The same-mark generator reorderings in the joint affine representation leave endpoint spans unchanged, so this remains true after reordering. For fixed J, Z, the second retained movable set is $Z \cup J'$ for some $J' \subseteq J$. The reconstruction lemma gives at most 2^j second bases, where $j = |J|$, with no factor depending on $z = |Z|$. Second-stream generators are not all forced; the finite checks include omitted ones.

Keep the three generator types and the same $j + z$ variables from the joint affine representation. On any fixed set of m tested labels, its $2m$ affine forms have at most $S_{j+z}(2m)$ strict sign patterns. Consequently, for fixed J, Z, the residual count is at most

$$6^j S_{j+z}(2m).$$

This counts the restricted streams having the forced-retention properties; every actual stream belongs to this family. Old paths, retained bases for each subset, and affine coefficients are fixed before the focal masks. Timeline length affects coefficients, not dimension. All three generator types, including early invalid insertions, remain charged.

8.2 Project the witnesses and residual onto the comparison optimum

Fix a phase and filter H, and let I^* be the fixed tie-broken optimum of the rounded full vector. Put $Y = I^* \setminus H$. For a true group U, write $n = |U|$ and $y = |Y \cap U|$. Condition further on parity and masks outside U, without conditioning on listing or any focal bits. Both U and $Y \cap U$ are fixed. The online rule does not receive or compute I^*.

At each valid nominal birth layer choose, by fixed label order, a basis extension

$$Z'_b \subseteq (P_b \cap Y) \setminus O \quad \text{over } X(b-2) \cup K_b, \qquad O = D \cup C.$$

Let $z' = \sum_b |Z'_b|$ and

$$R_Y = \bigcup_b \left[(P_b \cap Y) \setminus \mathrm{cl}(X(b-2) \cup K_b \cup Z'_b) \right].$$

The layers are disjoint, so $Z' = \bigcup_b Z'_b$ has size z' and is contained in $Y \cap U$. Its spanning property gives $R_Y \subseteq O$. All safe samples counted in σ lie in Y. A safe sample at layer b outside R_Y lies in the witness span but outside $Q'_b = Q(F(b-2) \cup K_b)$. Since $X(b-2) \cup K_b \subseteq Q'_b$, the source density inequality bounds their number by $\kappa\, \mathrm{r}(Z'_b \mid Q'_b) \le \kappa |Z'_b|$. Thus

$$\sigma \le \kappa z' + |R_Y|.$$

The full nominal statistic satisfies $z \geq z'$ by rank monotonicity in each layer. Lower-bounding z' therefore lower-bounds the full z in the original test-thinning lemma. The alternative witnesses use no focal test bits.

Use Z'_b in the two streams and test only $Y \cap U$. Their mark-one intersection is exactly R_Y, and both forced-retention properties continue to hold. Summing over generator and witness subsets gives fixed families

$$|\mathcal{F}_{j,z'}| \leq \binom{n}{j}\binom{y}{z'}6^j S_{j+z'}(2y).$$

Impossible subset choices only overcount. Generators may overlap the tested labels; no independence involving their adaptive selection is used. This bound includes zero sizes and dimensions exceeding $2y$.

In the refinement below, write z for the projected statistic z' just defined, and z_{full} for the original nominal statistic. For a true group U, put $n = |U|$ and $y = |Y \cap U|$.

8.3 Keep activation loss conditional on activation generators

Fix the rounded full vector, H, D, and a listed group. Put $J_0 = \{g \in J : \tau(g) = a\}$ and $J_1 = J \setminus J_0$, and write their sizes as j_0, j_1. The canonical forward construction gives

$$j_0 = \mathrm{r}(Q(F_\varnothing)), \qquad \tau(g) \geq a + 2 \ (g \in J_1), \qquad \kappa(j_0 + j_1) \leq |D \cap U|.$$

There are no generators at $a+1$, because the nominal flat does not change. Write $Y = I^* \setminus H$, as before. Membership in $Q(F_\varnothing)$ is determined by H, D and the fixed weights, before any guard or test bits are drawn. For each sampled $d \in Y \cap U$ outside that flat, every possible birth is valid (before the parity restriction). Apply the quadratic exit bound to this fixed label. Its squared-exit mass is at least $1/2$, all at valid births; the hybrid implication and independent parity therefore give safety probability at least $1/4$. Sum only over these labels. Since Y is independent, at most j_0 of its sampled labels lie in $Q(F_\varnothing)$. Thus

$$\mathbb{E}[\sigma \mid H, D] \geq \frac{|D \cap Y \cap U| - |D \cap Y \cap U \cap Q(F_\varnothing)|}{4} \geq \frac{|D \cap Y \cap U| - j_0}{4}.$$

This does not subtract a unit exit contribution from an unconditional half-exit bound. It applies the half-exit bound directly to the labels with no excluded birth. On unlisted groups all these statistics are zero. On averaging D, for a true group with $y = |Y \cap U|$,

$$\mathbb{E}\sigma \geq \rho y/4 - \mathbb{E}j_0/4.$$

8.4 Activation generators require no time variables

Fix H, parity, and all masks outside a true group U; its potential listed index and activation a are fixed, without conditioning on listing. Fix disjoint label sets J_0, J_1, Z, where Z is the projected witness set of the preceding proof. Every generator belongs to D, every witness belongs to $Y \setminus (D \cup C)$, and all these role sets are disjoint.

In both marked streams, J_0 has a fixed insertion position and fixed scores. In the first stream its time is a, with negative mark. In the second it is inserted at endpoint a or $a+1$, depending on parity, with positive mark. Both endpoints are invalid births; there are no competitors or witnesses there, and their incoming old flat is the loop flat. Thus these occurrences are retained. Incorporate

them into the fixed old lists. Their labels are still enumerated; only their insertion-time variables and type choices disappear.

All active generators are retained in the first stream. Old-basis reconstruction therefore fixes that retained basis for the specified sets. In the second stream all witnesses are retained, and an arbitrary subset of the j_1 active generators can be retained. There are at most 2^{j_1} second bases. It would be invalid to force every active generator there.

Because every active time is at least $a+2$, the shared affine representation has only two active-generator types: $(1,1)$ for a valid-parity insertion and $(-1,2^T)$ for the preceding-parity insertion shifted to its endpoint. The earlier third type was an activation occurrence, now a fixed constant. All fixed coefficients depend only on the enumerated role sets and the outside conditioning. There are j_1+z common variables, with $z=|Z|$, and at most $2y$ affine forms on the tested labels $Y\cap U$. For each fixed role assignment the number of residuals is at most

$$4^{j_1}S_{j_1+z}(2y), \qquad S_d(m)=\sum_{i=0}^{\min(d,m)}\binom{m}{i}.$$

This counts all actual streams before exposing the focal bits. It includes zero sizes, loops in old lists, repeated old occurrences, and dimensions exceeding $2y$. The old reconstruction and affine proofs apply with the fixed activation occurrences added to their old lists.

8.5 Use the joint mask-membership event

Let $O=D\cup C$, $p=\rho+(1-\rho)t$, and $\lambda=-\log p$. For each fixed role assignment and fixed candidate residual $R\subseteq Y\setminus Z$, put $J=J_0\cup J_1$. Across labels the focal mask pairs are independent, so the exact joint probability is

$$\Pr(J\subseteq D,\ Z\cap O=\varnothing,\ R\subseteq O)=\rho^{|J|}(1-p)^{|Z|}p^{|R\setminus J|}\le\rho^{|J|}p^{-|J\cap Y|}(1-p)^{|Z|}p^{|R|}.$$

The correction $p^{-|J\cap Y|}$ is essential: a generator can also be in the residual, so multiplying independent-looking generator and residual probabilities would undercount. Witnesses are disjoint from both the generators and the residual; invalid assignments can be omitted.

Choose $\mu>1/4$, $\nu>0$, and positive tilts $u,w,v<1$ satisfying

$$\rho\left(p^{\mu-1/4}+\frac{4p^{\mu}}{v}\right)\le u, \qquad \frac{(1-p)p^{\nu}}{v}\le w.$$

On listed outcomes let $e=(|R|-(\mu-1/4)j_0-\mu j_1-\nu z)_+$; otherwise let $e=0$. Use the residual count, the joint probability bound, and $v^dS_d(2y)\le(1+v)^{2y}$ in a union bound for $e>x$, with $x\ge 0$. For labels outside Y, the disjoint roles are none, activation generator, and active generator. For labels in Y, there is also the witness role, and generator roles pay the factor $1/p$. Their generating products give

$$\Pr(e>x)\le\min\{1,Mp^x\}, \qquad M\le(1+u)^{n-y}(1+u/p+w)^y(1+v)^{2y}.$$

To see the generating weights explicitly, an activation generator has weight $\rho p^{\mu-1/4}$, an active generator $4\rho p^{\mu}/v$, and a witness $(1-p)p^{\nu}/v$. The first two sum to at most u, and the last is at most w. Enumerating these disjoint choices per label proves the displayed tail bound. The actual selected sets may depend on the whole mask; only each fixed choice uses the joint probability formula. No conditional independence of the selected residual is assumed.

Writing

$$A = \log(1+u), \qquad B_Y = \log\frac{1+u/p+w}{1+u} + 2\log(1+v),$$

integration gives $\mathbb{E}e \le (nA + yB_Y + 1)/\lambda$. The additive constant remains and will be paid by the maximum branch. For any $0 < d \le \lambda$, use $p \ge \rho$ and $\log(1+x) \le x$ to obtain the convenient rational bounds

$$\mathbb{E}e \le \frac{nu + yb_Y + 1}{d}, \qquad b_Y = \frac{u(1/\rho - 1) + w}{1+u} + 2v.$$

8.6 Cancel activation loss before averaging the generator budget

The projected residual comparison is still $\sigma \le \kappa z + |R|$. By the definition of e, pointwise

$$\sigma + j_0/4 \le (\kappa+\nu)z + \mu(j_0 + j_1) + e.$$

Take expectations and combine the safety bound, the integrated tail, and the generator budget. The $\mathbb{E}j_0/4$ terms cancel. The full nominal rank is at least the projected witness rank, so it satisfies the existing rank interface with

$$K = \kappa + \nu, \qquad \bar\gamma = \rho/4 - b_Y/d, \qquad \epsilon = \mu\rho/\kappa + u/d, \qquad a_0 = 1/d,$$
$$\mathbb{E}[z_{\text{full}} \mid H] \ge \frac{\bar\gamma y - \epsilon n - a_0}{K}.$$

The old explicit $\rho/(2\kappa)$ error has disappeared because the activation penalty is smaller by $1/4$, not because activation births are assumed safe. All projected witnesses still avoid focal test bits.

The preceding full payoff calculation applies unchanged after substituting the new rank interface. In particular retain both later-cost factors and define

$$L_0 = \epsilon + Kt + \frac{Kt + K\rho/\kappa}{B-1}, \qquad s = (1-h)(\bar\gamma - L_0/h).$$

Small groups and the additive error cost at most $t(\bar\gamma\kappa + a_0)m_*/K$. Verify $(1-t)(\bar\gamma\kappa + a_0)/K \le 1/4$, so the maximum branch cancels them. The phase average then gives

$$C = \frac{(1-t)\beta^2 ts}{L(\theta-1)K}, \qquad \beta = (B-1)/B.$$

All order minima remain inside seed expectations. The same lossless sample/value simulation covers the adversary's knowledge of ignored samples, sacrificed values, and unused seed bits.

8.7 Exact parameters, resources, and limitations

The algorithmic parameters are

$$\kappa = 176, \quad \rho = 3/8, \quad h = 3/4, \quad t = 11/2^{17},$$
$$\theta = 261/256, \quad L = 256, \quad B = \theta^L, \quad p_{\text{main}} = 1 - t.$$

Choose

$$\mu = 113/8, \quad \mu_0 = 111/8, \quad \nu = 117/8, \quad K = 1525/8,$$
$$u = 1/1056, \quad w = 1/4096, \quad v = 1/648, \quad d = 49/50.$$

The exact certificate checks $\lambda > d$ by outward log intervals and a separate 20-term positive atanh sum. Since $p > (3/4)^4$, we have $p^{-1/4} < 4/3$. Set $A_* = \rho(4 + 4v/3)$. The separate exact power inequalities

$$A_*^8 p^{113} < (uv)^8, \qquad (1-p)^8 p^{117} < (wv)^8$$

prove both tilt inequalities; log intervals independently cross-check these finite scalar comparisons. The same certificate verifies $B > 140$, $L(\theta - 1) = 5$, positive slack, maximum-branch cancellation, and

$$\bar{\gamma} = \frac{23826973}{268494912}, \qquad \epsilon = \frac{51433}{1655808}, \qquad a_0 = \frac{50}{49},$$
$$s > \frac{9225899434547}{1681525638955008} > 0.$$

Using $\beta > 139/140$ and $B - 1 > 139$ in the final coefficient yields $C > C_4$. It also reruns the preceding certificate before comparing the improvement, and checks that $C_4 < 2^{-30}$. No claim is made to cross that next threshold.

The resulting conservative coefficient is

$$C_4 = \frac{8003458533570087953}{16807475946150704657203200000} > \frac{51}{50}\, 2^{-31} > 2^{-31}.$$

This is approximately $4.76184440734 \times 10^{-10}$, or 1.14973122069 times the preceding exact C_3: a 14.97% increase in the guaranteed fraction. It is approximately 31.44938 times the initial announced C_0 certificate above 2^{-36}. These compare guarantees, not measured performance.

The fair-bit implementation uses $2+3+17+17+1 = 40$ bits per label and $8+17+1 = 26$ shared bits, all drawn initially. The phase distribution is still uniform on 256 rational offsets. No additional sample, access to the comparison optimum, or bounded-precision assumption is introduced.

The numerical search executes 24 deterministic starts in the previous formula and each of three cumulative modes: conditional safety, activation splitting, and joint membership. Its selected dyadic certificate is conservative; the search is not a global optimum proof. The finite checker checks activation-fixed affine streams and conditional safety over all filter masks on the inherited small matroid/weight cases, exact joint membership laws with overlaps, disjoint-role generating products, and adaptive family selection. These finite facts do not prove the universal assertions above.

See the same-agent audit and baseline refresh. Independent review and formalization remain pending. No priority or confirmed strongest-known-bound claim is made.

9 Sources, attribution, and comparison limits

OpenAI supplies the original qualitative constant guarantee and construction in *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary*, September 23, 2026, pinned at revision fd4aeeb2ee4f. Its density paths, reverse exposure, hybrid comparison, marked-pivot machinery, rank accounting, and lossless sample/value simulation are fundamental inputs. The manuscript derives 2^{-293}, stronger than its 2^{-310} headline. The secretary consequence is in the same manuscript. Primary sources and hashes are recorded in the baseline.

This project's quadratic potential, joint affine encoding, and application of weighted residual counting refine that construction's quantitative analysis. The current version reconstructs old retained bases by contraction, uses forced retained occurrences, projects witnesses onto the

comparison optimum, and pays the tail's additive constant through the maximum branch. It further conditions safety on activation membership, assigns fixed scores to activation generators, and uses the joint generator/witness/residual mask law with its generator-residual overlap correction. Greedy scanning in a contracted matroid is standard; the claimed refinement is its application and the resulting quantitative accounting here. Exponential weighting, union bounds, finite-offset rounding, and binomial estimates are established methods; no new general principle is claimed for them. Each version retains its specific evidence and source credit in the result registry and archives.

Daniel (@firsttimelifer) announced a 2^{-47} guarantee on October 8, 2026. A supporting manuscript was not located in the October 10 audit; its assumptions and proof remain unverified. It is acknowledged as an announcement, not an established comparison baseline. The 1/2 prophet result in Abdi, Banihashem, Hajiaghayi, and Mittal, arXiv:2609.19118v1 assumes a fixed order independent of samples and values and therefore uses a different adversary model.

This work was developed with Codex assistance. The paper's author is Ryan Shea. The written arguments and exact programs have same-agent verification; independent mathematical review and formalization of these refinements remain pending. No exhaustive priority search, confirmed best-known-bound claim, or practical performance claim is made. The repository uses Apache 2.0, with source attribution in NOTICE.

10 Development history

Result	Guarantee	Change	Review stage
Parameter refinement	$> 2^{-63}$	Refined source parameters, affine branch mixing, and finite rounding offsets.	Written proof; independent review pending
Logarithmic hazard potential	$> 2^{-43}$	Introduced a logarithmic hazard potential and tighter residual accounting.	Written proof; independent review pending
Quadratic hazard potential	$> 2^{-36}$	A quadratic exit potential, integrated residual tail, and revised masks and finite phases.	Written proof; independent review pending
Joint residual encoding	$> 2^{-34}$	Shared generator variables across two pivot tests and charged a density-sample cutoff.	Written proof; independent review pending

Result	Guarantee	Change	Review stage
Rank-charged residual entropy	$> 2^{-32}$	The refinement weights residual families by generator and witness counts, charges their entropy to expected sample size and nominal rank, and removes the density cutoff.	Written proof; independent review pending
Retained bases and projected residuals	$> \frac{7}{4}\, 2^{-32}$	Retained-basis reconstruction, projected witnesses, and an additive maximum-branch charge improve the guarantee.	Written proof; independent review pending
Activation generators and joint mask membership	$> \frac{51}{50}\, 2^{-31}$	Conditional activation safety, fixed activation scores, and joint mask membership raise the guarantee above 2^{-31}.	Written proof; independent review pending

Earlier results retain their original evidence and source credit. These are stages of one research effort, not independent priority claims. Preserved publications are indexed in the archive.